

Calculus 2

5 Lecture 5

Functional Sequences and Series

Definition 5.1. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$.

Then:

1. the sequence $\{f_n(x)\}_{n=1}^{\infty}$ is called a *functional sequence*;
2. the series $\sum_{n=1}^{\infty} f_n(x)$ is called a *functional series*.

Definition 5.2. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$.

Then:

1. the sequence $\{f_n(x)\}_{n=1}^{\infty}$ *converges pointwise* to a function $f : E \rightarrow \mathbb{R}$ (notation: $f_n \xrightarrow{E} f$) on the set E if

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \quad \text{for every } x \in E.$$

That is,

$$(f_n \xrightarrow{E} f) \iff (\forall x \in E \ \forall \varepsilon > 0 \ \exists N(\varepsilon, x) \in \mathbb{N} \ \forall n \geq N(\varepsilon, x) \rightarrow |f_n(x) - f(x)| < \varepsilon). \quad (1)$$

2. the functional series $\sum_{k=1}^{\infty} f_k(x)$ *converges pointwise* to a function $f : E \rightarrow \mathbb{R}$ (notation: $\sum_{k=1}^{\infty} f_k \xrightarrow{E} f$) on the set E if the sequence of partial sums

$$\left\{ S_n(x) = \sum_{k=1}^n f_k(x) \right\}_{n=1}^{\infty}$$

converges pointwise to f .

Claim 5.1 (Properties of Finite Sums). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

Then:

1. if every u_k is continuous on E , then f_n is also continuous on E ;
2. if every u_k is differentiable on E , then f_n is also differentiable on E and

$$f'_n(x) = \sum_{k=1}^n u'_k(x), \quad \forall x \in E;$$

3. if every u_k is Riemann integrable on $[a, b] \subseteq E$, then f_n is also Riemann integrable on $[a, b]$ and

$$\int_a^b f_n(x) dx = \sum_{k=1}^n \int_a^b u_k(x) dx.$$

Proof. See Calculus-1. □

Can we extend Properties 1), 2), and 3) from [Claim 5.1](#) to the series

$$f(x) = \sum_{k=1}^{\infty} u_k(x)?$$

In general, **NO** (see the next Claim), but **YES** under some assumptions.

The main goal of this and the next lectures is to establish such assumptions.

Claim 5.2. Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

Then:

1. If every u_k is continuous on E and $f_n \xrightarrow{E} f$, then f may not be continuous on E .
2. If every u_k is differentiable on E and $f_n \xrightarrow{E} f$, then f may not be differentiable on E .
3. If every u_k is Riemann integrable on $[a, b] \subseteq E$ and $f_n \xrightarrow{E} f$, then f may not be Riemann integrable on $[a, b] \subseteq E$.

Note that, [Claim 5.2](#) shows that the assumption $f_n \xrightarrow{E} f$ is not strong enough, if we want to extend the properties (i.e. continuity, differentiability and Riemann integrability) to

$$f(x) = \sum_{k=1}^{\infty} u_k(x),$$

we need to replace the assumption $f_n \xrightarrow{E} f$ with a stronger one. We will prove the corresponding claims in the next lecture.

Proof. 1) and 2). Let $E = \mathbb{R}$,

$$u_k(x) = \frac{\sin(kx)}{k}, \quad \forall k \in \mathbb{N}.$$

Then

$$\sum_{k=1}^{\infty} \frac{\sin(kx)}{k} \xrightarrow{\mathbb{R}} f(x),$$

that is, the series converges pointwise on $\mathbb{R}$ (due to [Claim 3.2](#) [Dirichlet's Test]).

It can be shown that (we will be able to do it later in our course) that the function f is of the form:

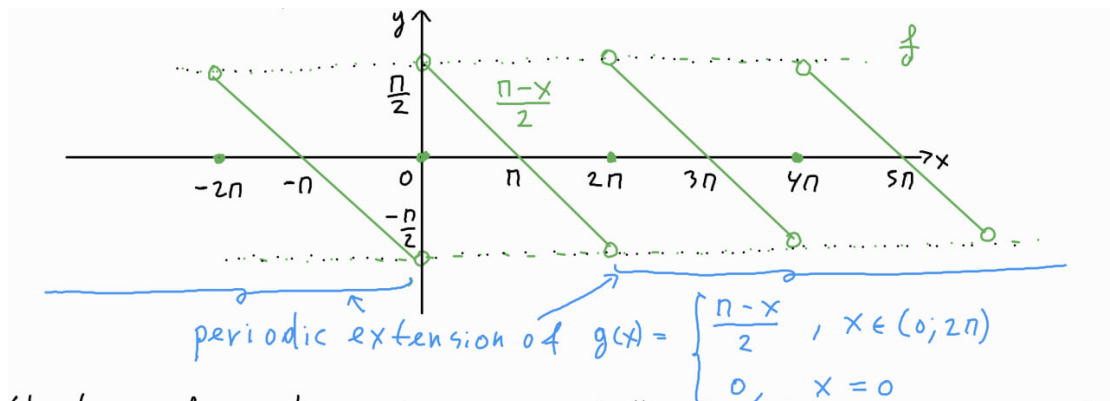


Figure 5.1

It is the periodic extension of

$$g(x) = \begin{cases} \frac{\pi - x}{2}, & x \in (0, 2\pi), \\ 0, & x = 0. \end{cases}$$

That is, f is discontinuous at the points $x = 2\pi m$, $m \in \mathbb{Z}$.

Since, for every $k \in \mathbb{N}$,

$$u_k(x) = \frac{\sin(kx)}{k}$$

is infinitely differentiable on $\mathbb{R}$, [Items 1\)](#) and [2\)](#) are proved.

Also note that the equality

$$f'(x) = \sum_{k=1}^{\infty} \left(\frac{\sin(kx)}{k} \right)' = \sum_{k=1}^{\infty} \cos(kx)$$

has no sense (since the latter series diverges for every $x \in \mathbb{R}$).

3) Let $E = [0, 1]$, $\tilde{\tau} : \mathbb{N} \rightarrow [0, 1] \cap \mathbb{Q}$ be a bijection (you should know that such a bijection exists; why?) and

$$u_k(x) = \begin{cases} 1, & x = \tilde{\tau}(k), \\ 0, & x \in [0, 1] \setminus \{\tilde{\tau}(k)\}, \end{cases} \quad \forall k \in \mathbb{N}.$$

Then,

$$\sum_{k=1}^{\infty} u_k(x) \xrightarrow{E} f(x) = \begin{cases} 1, & x \in [0, 1] \cap \mathbb{Q}, \\ 0, & x \in [0, 1] \text{ and } x \notin \mathbb{Q}. \end{cases}$$

That is, f is the Dirichlet function on $[0, 1]$.

It ought to be clear that f is not Riemann-integrable on $E = [0, 1]$ (since f is nowhere continuous on $[0, 1]$). Since every u_k has finitely many points of discontinuities (actually just one point: $x = \tilde{\tau}(k)$) and it is bounded on $[0, 1]$, it is Riemann-integrable on $[0, 1]$. Thus, the item is proved. $\square$

Definition 5.3. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$.

Then:

1. the functional sequence $\{f_n(x)\}_{n=1}^{\infty}$ is called *uniformly convergent* to a function $f(x)$ on a set E (notation $f_n \xrightarrow{E} f$) if

$$\lim_{n \rightarrow \infty} \sup_{x \in E} |f_n(x) - f(x)| = 0.$$

That is,

$$(f_n \xrightarrow{E} f) \iff (\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \varepsilon). \quad (2)$$

2. the functional series $\sum_{k=1}^{\infty} f_k(x)$ is called *uniformly convergent* to a function $f(x)$ on a set E (notation $\sum_{k=1}^{\infty} f_k(x) \xrightarrow{E} f(x)$) if the sequence of its partial sums

$$\left\{ S_n(x) = \sum_{k=1}^n f_k(x) \right\}_{n=1}^{\infty}$$

converges uniformly to $f(x)$ on E .

[Definition 2](#)) means that, in order to converge uniformly on E , the functions $f_n(x)$ ought to be inside the ε -neighbourhood of $f(x)$ for all $n \geq N(\varepsilon)$ and for all $x \in E$.

Claim 5.3. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$. Then

$$(f_n \xrightarrow{E} f) \implies (f_n \xrightarrow{E} f).$$

That is, if the sequence $\{f_n(x)\}_{n=1}^{\infty}$ converges uniformly to $f(x)$ on E , then it converges pointwise to $f(x)$ on E .

Note that the converse does not hold true in general.

Proof. ($\implies$) The statement follows directly from [Def 5.2](#) and [Def 5.3](#):

$$\begin{aligned} & (f_n \xrightarrow{E} f) \\ & \Updownarrow \quad \text{Def 5.3} \\ & (\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \varepsilon) \\ & \Downarrow \quad \text{since we can use } N(\varepsilon, x) = N(\varepsilon) \\ & (\forall \varepsilon > 0 \forall x \in E \exists N(\varepsilon, x) \in \mathbb{N} : \forall n \geq N(\varepsilon, x) \rightarrow |f_n(x) - f(x)| < \varepsilon) \\ & \Updownarrow \quad \text{Def 5.2} \\ & (f_n \xrightarrow{E} f). \end{aligned}$$

For the converse, let

$$f_n(x) = \begin{cases} 1, & x \in [n, n+1], \\ 0, & x \in \mathbb{R} \setminus [n, n+1], \end{cases} \quad f(x) = 0 \quad \forall x \in \mathbb{R}, \quad E = \mathbb{R}.$$

Then $\forall x \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} f_n(x) = 0 = f(x).$$

That is, due to [Def 5.2](#), $f_n \xrightarrow{\mathbb{R}} f$. Since

$$\sup_{x \in \mathbb{R}} |f_n(x) - f(x)| = 1,$$

we have

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| = 1 \neq 0.$$

Therefore, due to [Def 5.3](#), $f_n \not\xrightarrow{\mathbb{R}} f$. □

Claim 5.4 (The Cauchy Criterion for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$. Then

$$\begin{aligned} & (f_n \xrightarrow{E} f) \\ & \Updownarrow \\ & (\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_k(x)| < \varepsilon). \end{aligned} \quad (3)$$

Equivalently,

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall p \in \mathbb{N} \forall x \in E \rightarrow |f_{m+p}(x) - f_m(x)| < \varepsilon).$$

Proof. ($\Downarrow$) Note that

$$(f_n \xrightarrow{E} f)$$

by [Formula 2](#) implies

$$(\forall \varepsilon > 0 \exists N_1(\varepsilon) : \forall n \geq N_1(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{2}).$$

Let $N(\varepsilon) = N_1(\varepsilon)$ in [Formula \(3\)](#). Then, for $\forall k \geq N(\varepsilon)$, $\forall m \geq N(\varepsilon)$, $\forall x \in E$, we have

$$\begin{aligned} |f_m(x) - f_k(x)| &= |f_m(x) - f(x) + f(x) - f_k(x)| \\ &\leq |f_m(x) - f(x)| + |f_k(x) - f(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus, we proved ($\Downarrow$).

($\Uparrow$) Note that

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_k(x)| < \varepsilon)$$

implies

$$(\forall x \in E \forall \varepsilon > 0 \exists N(\varepsilon, x) \in \mathbb{N} : \forall k \geq N(\varepsilon, x) \forall m \geq N(\varepsilon, x) \rightarrow |f_m(x) - f_k(x)| < \varepsilon).$$

For a fixed $x \in E$, this is the Cauchy convergence criterion for the sequence $\{f_n(x)\}_{n=1}^{\infty}$ (see Calculus-1). Therefore,

$$\forall x \in E \rightarrow \exists \lim_{n \rightarrow \infty} f_n(x).$$

Therefore, we can define $f : E \rightarrow \mathbb{R}$ as

$$f(x) \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} f_n(x), \quad \forall x \in E.$$

Thus, we proved that $f_n \xrightarrow{E} f$.

Now, if in [Formula \(3\)](#) we take the limit $k \rightarrow \infty$, we will receive [Formula 2](#) from [Def 5.3](#):

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f(x)| \leq \varepsilon.$$

Here we used that $\forall x \in E \exists \lim_{k \rightarrow \infty} f_k(x) = f(x)$. Therefore, due to [Def 5.3](#), $f_n \xrightarrow{E} f$. $\square$

Claim 5.5 (The Negation of the Cauchy Criterion for the UC). Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$. Then

$$(f_n \not\xrightarrow{E} f)$$

$$\Uparrow$$

$$(\exists \varepsilon > 0 \forall N \in \mathbb{N} \exists k \geq N \exists m \geq N \exists x \in E \rightarrow |f_m(x) - f_k(x)| \geq \varepsilon). \quad (4)$$

Example. Study the uniform convergence of the sequence

$$\left\{ f_n(x) = \frac{\ln(nx)}{\sqrt{nx}} \right\}_{n=1}^{\infty}$$

on the set $E = (0, 1)$.

Solution. Since, for every $x \in (0, 1)$, we have

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{\ln(nx)}{\sqrt{nx}} = 0 \quad // \text{see Calculus-1} //$$

due to [Def 5.2](#), the sequence $\{f_n(x)\}_{n=1}^{\infty}$ converges pointwise to the function $f \equiv 0$ on $E = (0, 1)$.

Now, using [Claim 5.5](#), we will prove that $f_n \not\stackrel{E}{\rightrightarrows} f \equiv 0$. In [Formula \(4\)](#), let

$$\varepsilon = \frac{1}{4}, \quad k = N, \quad m = 2N, \quad x = \frac{1}{N} \in E = (0, 1).$$

Then, we have:

$$\begin{aligned} |f_m(x) - f_k(x)| &= \left| f_{2N}\left(\frac{1}{N}\right) - f_N\left(\frac{1}{N}\right) \right| \\ &= \frac{\ln(2N \cdot \frac{1}{N})}{\sqrt{2N \cdot \frac{1}{N}}} - \frac{\ln(N \cdot \frac{1}{N})}{\sqrt{N \cdot \frac{1}{N}}} \\ &= \frac{\ln 2}{\sqrt{2}} - \frac{\ln 1}{\sqrt{1}} = \frac{\ln 2}{\sqrt{2}} > \frac{1}{4} = \varepsilon. \end{aligned}$$

Thus, due to [Claim 5.5](#), $f_n \not\stackrel{E}{\rightrightarrows} f \equiv 0$.

Claim 5.6 (Necessary Condition for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad n \in \mathbb{N}.$$

Then

$$(f_n \stackrel{E}{\rightrightarrows} f) \implies (u_k \stackrel{E}{\rightrightarrows} 0).$$

That is, if the series $\sum_{k=1}^{\infty} u_k(x)$ converges uniformly on E , then the sequence $\{u_k(x)\}_{k=1}^{\infty}$ converges uniformly to the zero function.

Compare [Claim 5.6](#) with [Claim 1.5](#).

Proof. Note that

$$(f_n \stackrel{E}{\rightrightarrows} f)$$

due to [Claim 5.4](#), for $k = m - 1$, implies

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_{m-1}(x)| < \varepsilon).$$

Since

$$f_m(x) - f_{m-1}(x) = \sum_{k=1}^m u_k(x) - \sum_{k=1}^{m-1} u_k(x) = u_m(x),$$

this is equivalent to

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |u_m(x) - 0| < \varepsilon),$$

which, due to [Def 5.3](#) and [Formula \(2\)](#), is

$$u_m \stackrel{E}{\rightrightarrows} 0.$$

□

Note. The necessary condition ([Claim 5.6](#)) is not sufficient. It cannot even guarantee the pointwise convergence. Let $E = (0, 1)$, and

$$u_k(x) = \frac{1}{k+x}.$$

Then $u_k(x) \xrightarrow{E} 0$, but the series

$$\sum_{k=1}^{\infty} \frac{1}{k+x}$$

is divergent for every $x \in E = (0, 1)$.

Claim 5.7 (The Weierstrass M-test). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad n \in \mathbb{N}.$$

Suppose that there is a sequence of real numbers $\{M_k\}_{k=1}^{\infty}$ satisfying the conditions:

a) $\forall k \in \mathbb{N} \forall x \in E \rightarrow |u_k(x)| \leq M_k$;

b) $\sum_{k=1}^{\infty} M_k$ converges.

Then, the series

$$\sum_{k=1}^{\infty} u_k(x)$$

converges uniformly on E . That is, $f_n \xrightarrow{E} f$.

Proof. First, note that

$$\left| \sum_{k=m+1}^{m+p} M_k \right| < \varepsilon.$$

From [Item a](#)),

$$(\forall k \in \mathbb{N} \forall x \in E \rightarrow |u_k(x)| \leq M_k) \implies (\forall k \in \mathbb{N} \rightarrow M_k \geq 0).$$

Therefore,

$$\sum_{k=m+1}^{m+p} M_k < \varepsilon.$$

Using again $\forall k \in \mathbb{N} \forall x \in E \rightarrow |u_k(x)| \leq M_k$, we obtain

$$\forall x \in E \rightarrow \sum_{k=m+1}^{m+p} |u_k(x)| < \varepsilon,$$

and by

$$|a_1 + a_2 + \cdots + a_n| \leq |a_1| + |a_2| + \cdots + |a_n|, \quad \forall a_1, a_2, \dots, a_n \in \mathbb{R},$$

we get

$$\forall x \in E \rightarrow \left| \sum_{k=m+1}^{m+p} u_k(x) \right| < \varepsilon \iff \forall x \in E \rightarrow |f_{m+p}(x) - f_m(x)| < \varepsilon. \quad (**)$$

Due to [Claim 1.2](#), we have

$$\left(\sum_{k=1}^{\infty} M_k \text{ converges} \right)$$

(Item b)), hence

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall p \in \mathbb{N} \rightarrow \left| \sum_{k=m+1}^{m+p} M_k \right| < \varepsilon.$$

Due to Formula (**),

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall p \in \mathbb{N} \forall x \in E \rightarrow |f_{m+p}(x) - f_m(x)| < \varepsilon,$$

which is equivalent to Formula (3) from Claim 5.4. Therefore, due to Claim 5.4, $f_n \xrightarrow{E} f$. $\square$

Claim 5.8 (The Dirichlet Test for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, $a_k : E \rightarrow \mathbb{R}$, and $b_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$. Suppose that:

a) $\exists M > 0 : \forall n \in \mathbb{N} \forall x \in E \rightarrow \left| \sum_{k=1}^n a_k(x) \right| < M;$

that is, the sequence

$$\left\{ A_n = \sum_{k=1}^n a_k(x) \right\}_{n=1}^{\infty}$$

is uniformly bounded on E ;

b) either

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \geq b_{k+1}(x),$$

or

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \leq b_{k+1}(x);$$

c) $b_k \xrightarrow{E} 0$.

Then, the series

$$\sum_{k=1}^{\infty} a_k(x) b_k(x)$$

converges uniformly on E .

Claim 5.9 (The Abel Test for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, $a_k : E \rightarrow \mathbb{R}$, and $b_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$. Suppose that:

a) the series $\sum_{k=1}^{\infty} a_k(x)$ converges uniformly on E ;

b) either

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \geq b_{k+1}(x),$$

or

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \leq b_{k+1}(x);$$

c) $\exists M > 0 \forall k \in \mathbb{N} \forall x \in E \rightarrow |b_k(x)| \leq M;$

that is, the sequence $\{b_k(x)\}_{k=1}^{\infty}$ is uniformly bounded on E .

Then, the series

$$\sum_{k=1}^{\infty} a_k(x) b_k(x)$$

converges uniformly on E .

Claim 5.10. Let

$$S_n(x) = \sum_{k=1}^n \sin(kx) \quad \text{and} \quad C_n(x) = \sum_{k=1}^n \cos(kx), \quad \forall n \in \mathbb{N}.$$

Then:

1. $\forall \delta \in (0, \pi) \ \forall x \in [\delta, 2\pi - \delta] \ \forall n \in \mathbb{N} \rightarrow |S_n(x)| \leq \frac{1}{\sin(\delta/2)};$
2. $\forall \delta \in (0, \pi) \ \forall x \in [\delta, 2\pi - \delta] \ \forall n \in \mathbb{N} \rightarrow |C_n(x)| \leq \frac{1}{\sin(\delta/2)}.$

Proof. 1) Follows directly from [Claim 3.5](#) for $a = x$ and $b = 0$.

2) Follows directly from [Claim 3.5](#) for $a = x$ and $b = \frac{\pi}{2}$.

□